Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 02

Student ID	IT24101365
Student Name	Kamshiga Ganesan
Campus/ Center name	Northern UNI
Specialization	Software Engineering
Batch No	1

Activity 1

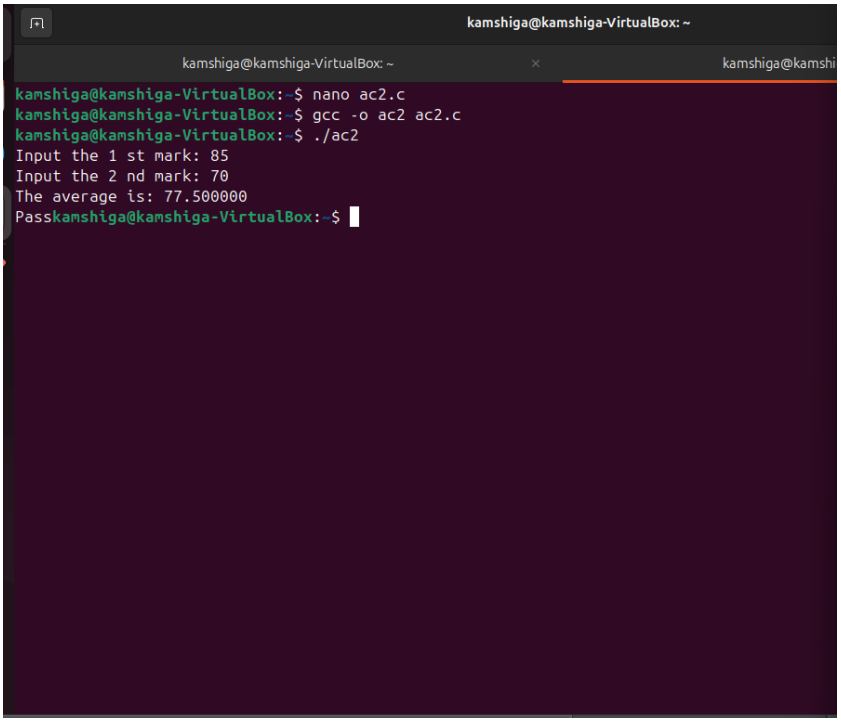
Create a new C program file and type the below program. Compile and execute it. Observe the result and explain it in your words.

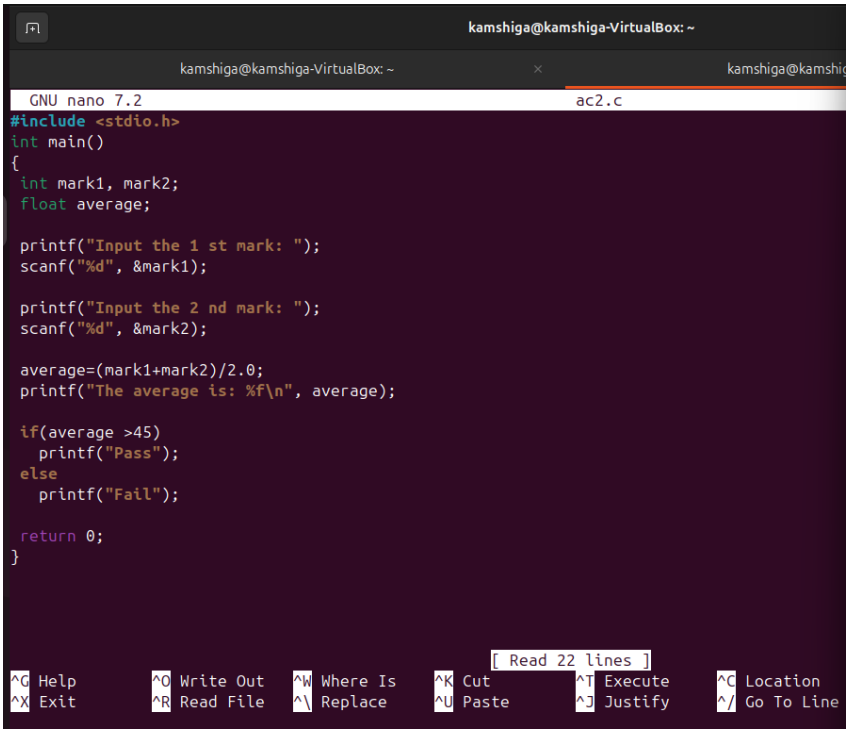
```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ac1.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ac1 ac1.c
kamshiga@kamshiga-VirtualBox:~$ ./ac1
Input the 1 st mark: 85
Input the 2 nd mark: 70
The average is: 77.500000
kamshiga@kamshiga-VirtualBox:~$
```

```
  GNU nano 7.2                                          ac1.c
#include <stdio.h>
int main()
{
 int mark1, mark2;
 float average;

 printf("Input the 1 st mark: ");
 scanf("%d", &mark1);

 printf("Input the 2 nd mark: ");
 scanf("%d", &mark2);

 average=(mark1+mark2)/2.0;
 printf("The average is: %f\n", average);

 return 0;
}

                                        [ Read 17 lines ]
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute    ^C Location
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line
```

Activity 2

Modify the above program to display ”Pass” if the average is above 45, otherwise display ”Fail”.





Activity 3

Write a C program to enter the gender and the age of a person from the keyboard. If the age is greater than or equal to 65, display ”Senior Male citizen” or ”Senior Female citizen” based on the gender

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ac3.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ac3 ac3.c
kamshiga@kamshiga-VirtualBox:~$ ./ac3
Enter gender(M/F): m
Enter age: 65
Senior Male citizenkamshiga@kamshiga-VirtualBox:~$ ./ac3
Enter gender(M/F): F
Enter age: 22
Not a senior citizenkamshiga@kamshiga-VirtualBox:~$ ./ac3
Enter gender(M/F): f
Enter age: 70
Senior Female citizenkamshiga@kamshiga-VirtualBox:~$
```

```
kamshiga@kamshiga-VirtualBox: ~
  GNU nano 7.2                                    ac3.c
#include <stdio.h>
int main()
{
 char gender;
 int age;

 printf("Enter gender(M/F): ");
 scanf("%c", &gender);

 printf("Enter age: ");
 scanf("%d", &age);

 if(age >=65){
  if(gender == 'M' || gender=='m')
   printf("Senior Male citizen");

  else if(gender=='F' || gender=='f')
   printf("Senior Female citizen");

  else
   printf("Invalide gender");
 } else {
   printf("Not a senior citizen");
 }

 return 0;

}

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line
```

Activity 4

Write a C program to enter the mark obtained for a module and display the grade based on the below criteria.

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ac4.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ac4 ac4.c
kamshiga@kamshiga-VirtualBox:~$ ./ac4
Enter the mark: 100
Grade Akamshiga@kamshiga-VirtualBox:~$ ./ac4
Enter the mark: 70
Grade Bkamshiga@kamshiga-VirtualBox:~$ ./ac4
Enter the mark: 69
Grade Ckamshiga@kamshiga-VirtualBox:~$ ./ac4
Enter the mark: 44
Grade Fkamshiga@kamshiga-VirtualBox:~$
```

```
kamshiga@kamshiga-VirtualBox: ~
  GNU nano 7.2                                    ac4.c
#include <stdio.h>
int main(){
 int mark;

 printf("Enter the mark: ");
 scanf("%d",&mark);

 if(mark<=100 && mark >=80)
  printf("Grade A");
 else if(mark<=79 && mark >=70)
  printf("Grade B");
 else if(mark<=69 && mark >=45)
  printf("Grade C");
 else
  printf("Grade F");

 return 0;
}

                                  [ Read 18 lines ]
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute    ^C Location
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line
```

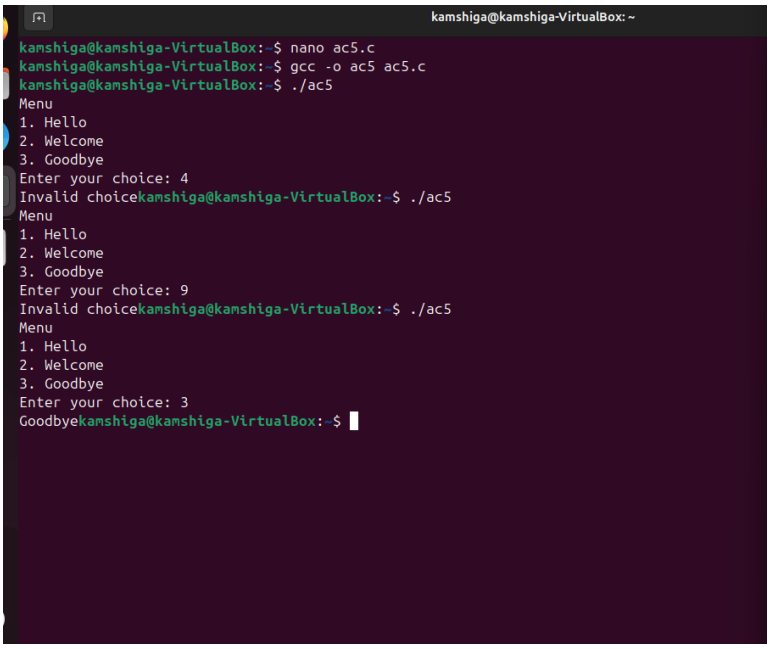
Activity 5

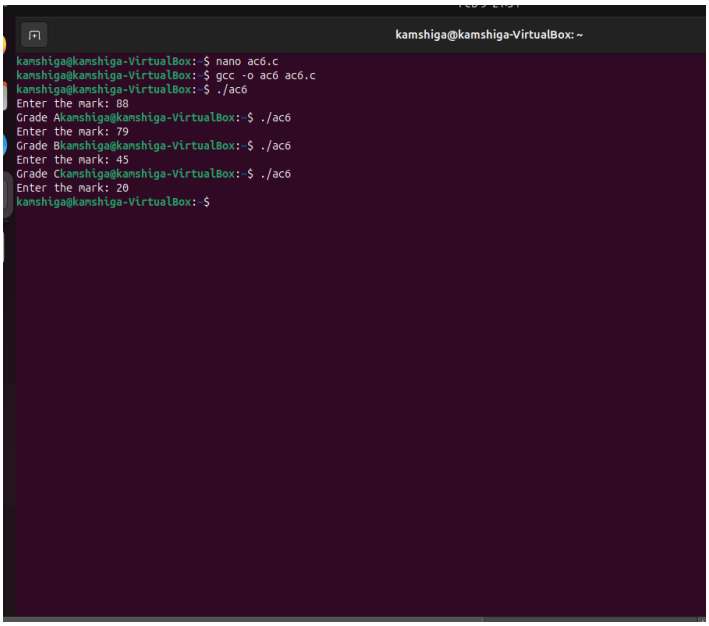
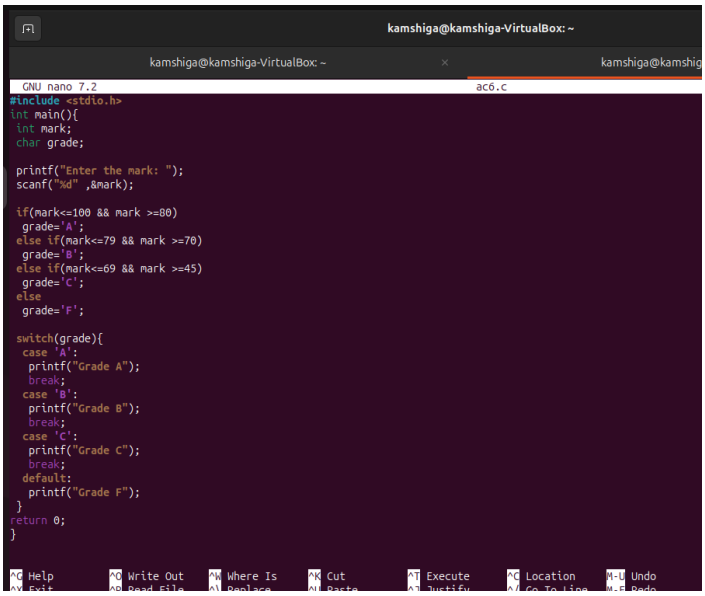
Write a C program that displays a menu with the following options:

1. Print ”Hello”

2. Print ”Welcome”

3. Print ”Goodbye” The program should ask the user to enter a choice (1 to 3) and use a switch-case state ment to display the corresponding message. If the user enters any other number, display ”Invalid choice”.



Activity 6

Consider the question given in activity 4. Rewrite the C program using switch– case statement. The mark of the module must be input into the program through the key board.